

Milestone Progress Report

This report summarizes the milestone progress for the project **MERN Stack Employee Management System**.

STUDENT NAME Bhanu Sharma	ENROLLMENT NUMBER O24MCA112292
COURSE MCA Full Stack Development	UNIVERSITY Chandigarh University
GUIDE / MENTOR Alok Srivastva	ACADEMIC YEAR 2024-2026

Progress Summary

6 COMPLETED	0 PENDING	100% OVERALL PROGRESS
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Milestone Status

MILESTONE	PROGRESS	ACTIONS / STATUS	WORK COMPLETED / EVIDENCE
Design Database Schema	COMPLETE	Marked as Complete	MongoDB schema designed using Mongoose models for users, employees, and activity logs. Employee records include name, email, phone, department, job title, status, creator, and timestamps.
Develop RESTful APIs	COMPLETE	Marked as Complete	Express APIs developed for authentication, employee CRUD, bulk CSV import, activity logs, and role-based user management.
Create Front-End Interface	COMPLETE	Marked as Complete	React interface completed with dashboard, employee list, employee details, add/edit employee, CSV import, departments, activity, profile, and role-based users pages.
Implement Authentication	COMPLETE	Marked as Complete	JWT authentication implemented with protected routes, login, registration, profile update, and admin, department-head, and user access.

MILESTONE	PROGRESS	ACTIONS / STATUS	WORK COMPLETED / EVIDENCE
Conduct Testing	COMPLETE	Marked as Complete	Build verification, npm audits, API checks, and Playwright smoke test completed. The Playwright test covers login, employee CRUD, search, details reload, departments, and CSV import validation.
Deploy Application	COMPLETE	Marked as Complete	Application deployed on Render with MongoDB Atlas as the production database. The backend serves the built React frontend from the same web service.

Current Status: All milestones are complete. The application is available locally for testing and deployed on Render for final submission.

Repository And Local Verification

GitHub Repository	https://github.com/nextlvlpro/mern-employee-app
Live Application	https://bluepeak-hr.onrender.com
Frontend URL	http://localhost:5173
Backend Health URL	http://localhost:5000/api/health
Testing Command	npm run test:e2e

Student Signature
Bhanu Sharma

Guide / Mentor
Alok Srivastva